

Functions & Graphs

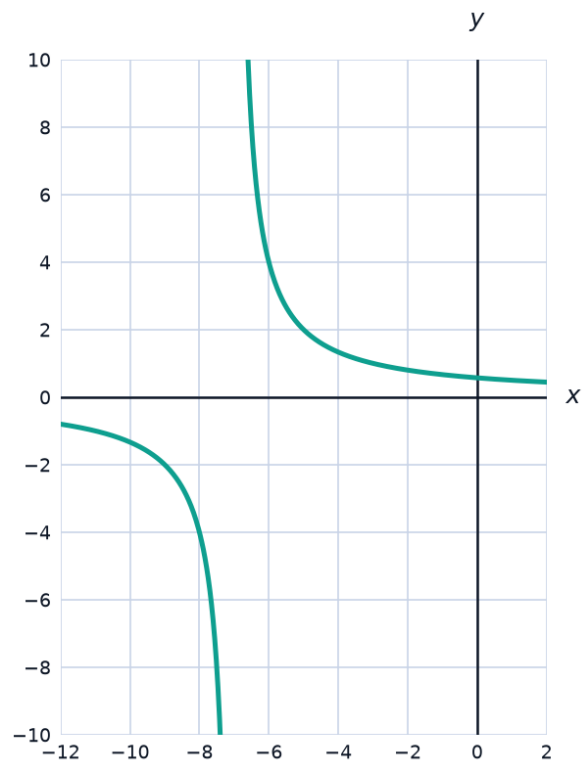


Evaluating functions

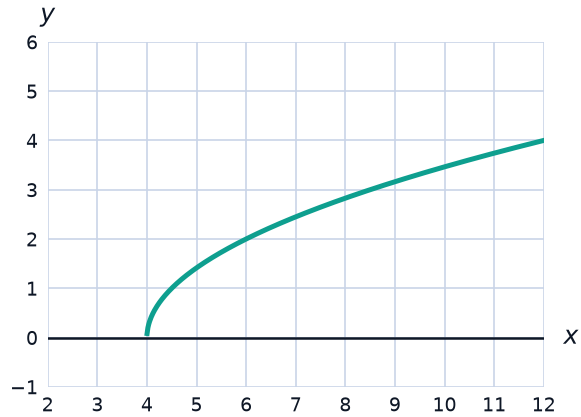
- 1 If $f(x) = 3x^2 - 2x + 5$, find $f(-2)$.
- 2 If $g(x) = \frac{2x + 1}{x - 3}$, find $g(5)$.

Domain and range

- 3 Find the domain of $f(x) = \frac{4}{x + 7}$.



- 4 Find the domain of $g(x) = \sqrt{2x - 8}$.

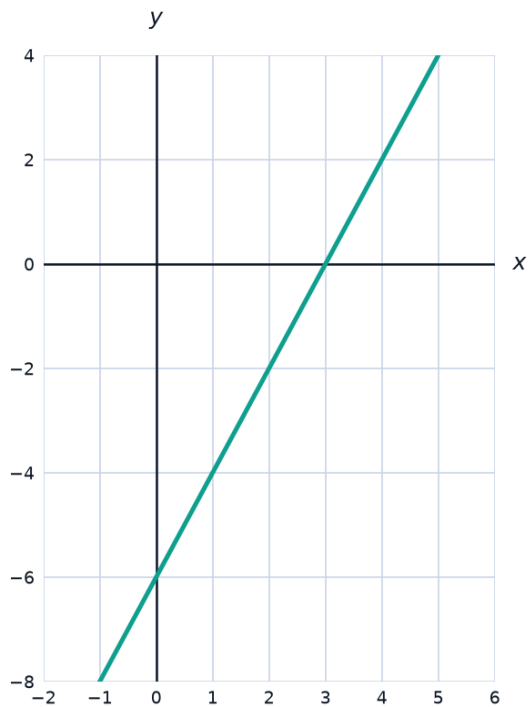


Composite functions

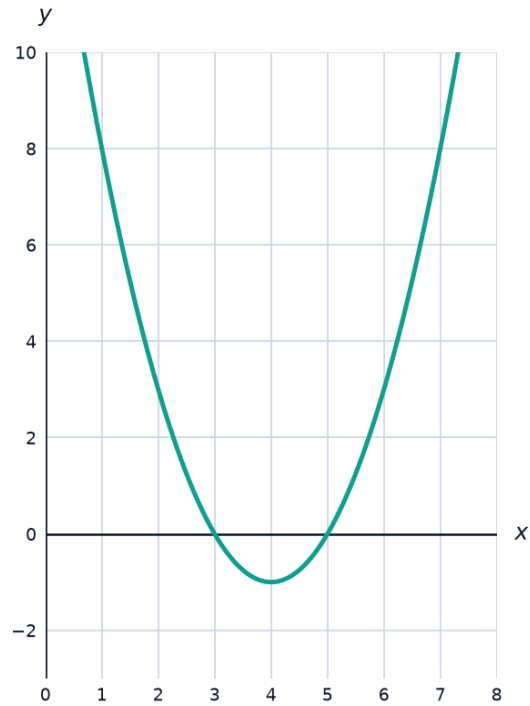
- 5 If $f(x) = 2x + 1$ and $g(x) = x^2 - 3$, find $f(g(2))$.
- 6 If $f(x) = x + 4$ and $g(x) = 3x$, find $(f \circ g)(x)$ and $(g \circ f)(x)$, and explain why they differ.

Graphing linear and quadratic functions

- 7 Find the x- and y-intercepts of $y = 2x - 6$.

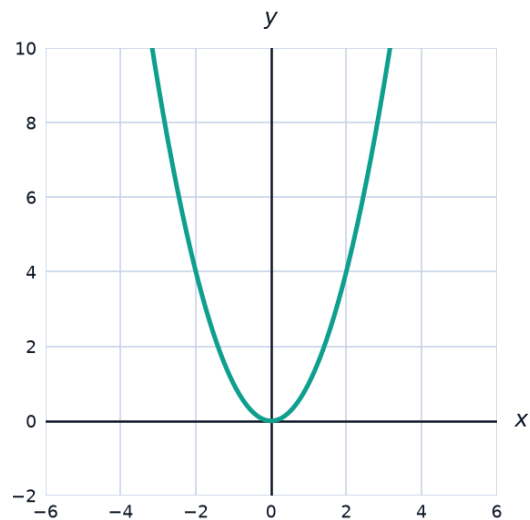


- 8 Find the vertex of $y = x^2 - 8x + 15$.

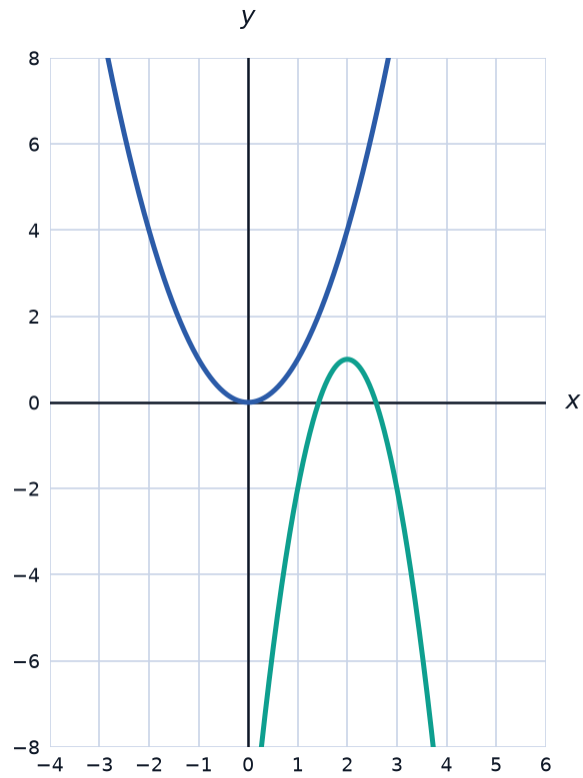


Transformations of graphs

- 9 The graph of $y = x^2$ is shifted 4 units left and 3 units down. Write the equation of the resulting graph.

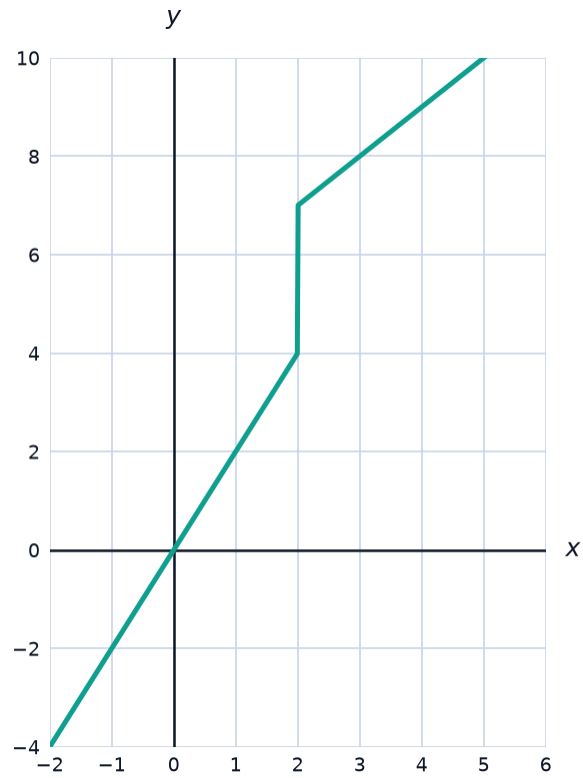


- 10 Describe how the graph of $y = -3(x - 2)^2 + 1$ differs from the graph of $y = x^2$.



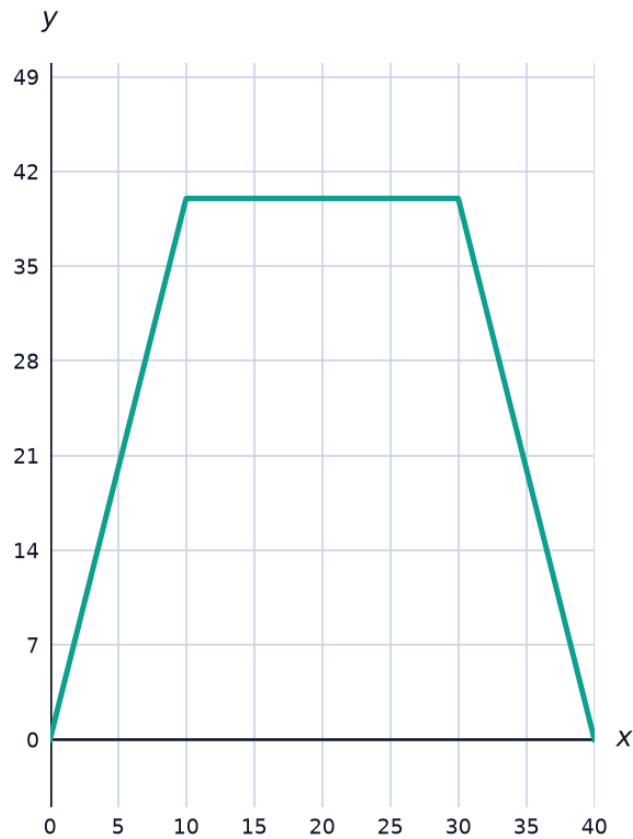
Piecewise and step functions

- 11 A function is defined as $f(x) = 2x$ for $x \leq 2$ and $f(x) = x + 5$ for $x > 2$. Find $f(0)$ and $f(4)$.



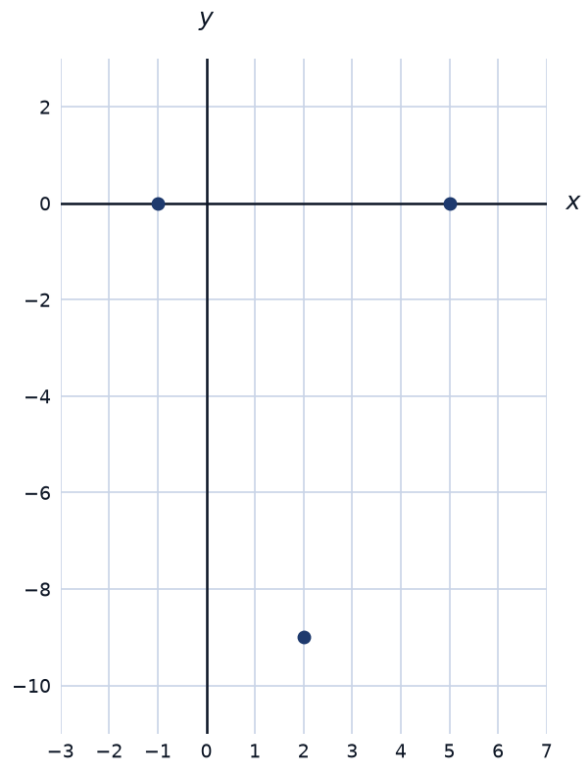
Interpreting graphs in context

- 12** A graph shows a car's speed over time: the speed increases from 0 to 40 mph over the first 10 seconds, stays constant at 40 mph for the next 20 seconds, then decreases to 0 over the final 10 seconds. Describe what is happening to the car during each phase.



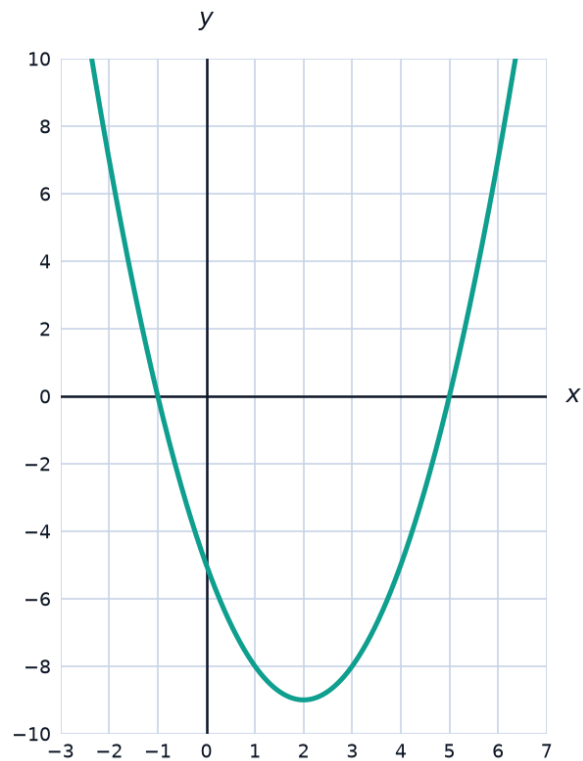
Finding the equation of a function from a graph description

- 13 A parabola has x -intercepts at $x = -1$ and $x = 5$, and passes through $(2, -9)$. Find its equation in factored form.



Multi-part function analysis

- 14** This question has two parts. A function is given by $f(x) = x^2 - 4x - 5$. (a) Find the zeros of the function by factoring. (b) Find the vertex, and state whether it is a minimum or maximum.

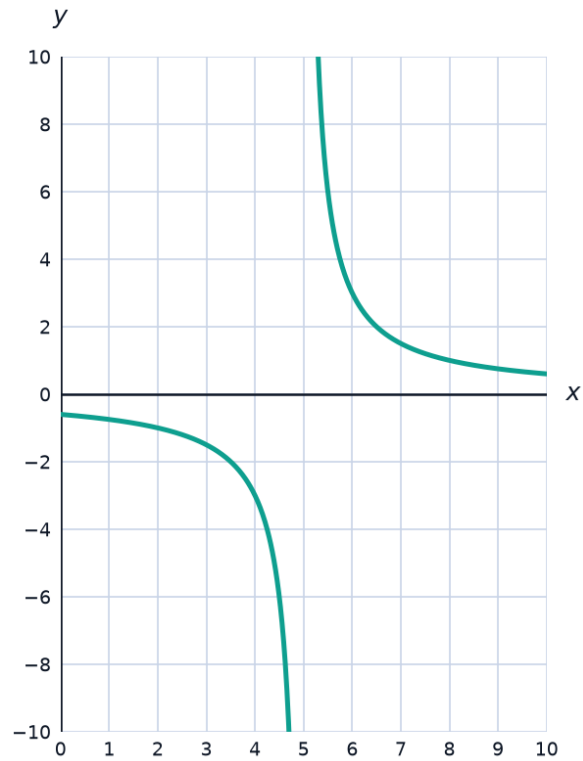


Inverse functions

- 15** Find the inverse of $f(x) = 4x + 8$.
- 16** If $f(3) = 10$, find $f^{-1}(10)$, using the definition of an inverse function.

Rational and radical functions

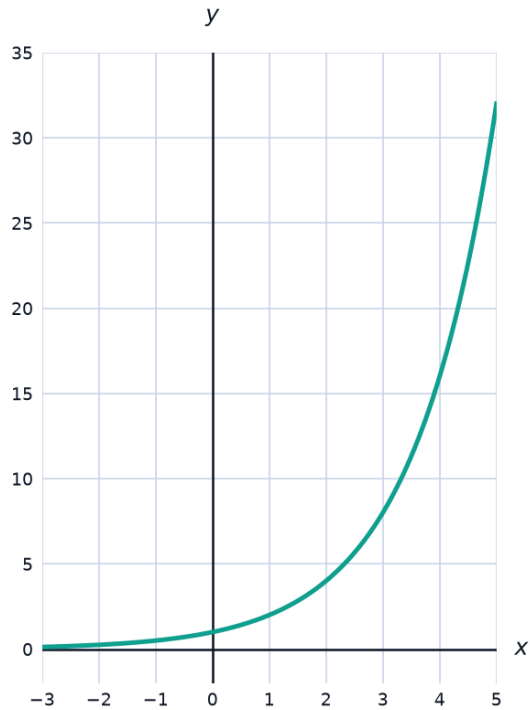
- 17 Find the vertical asymptote of $f(x) = \frac{3}{x-5}$.



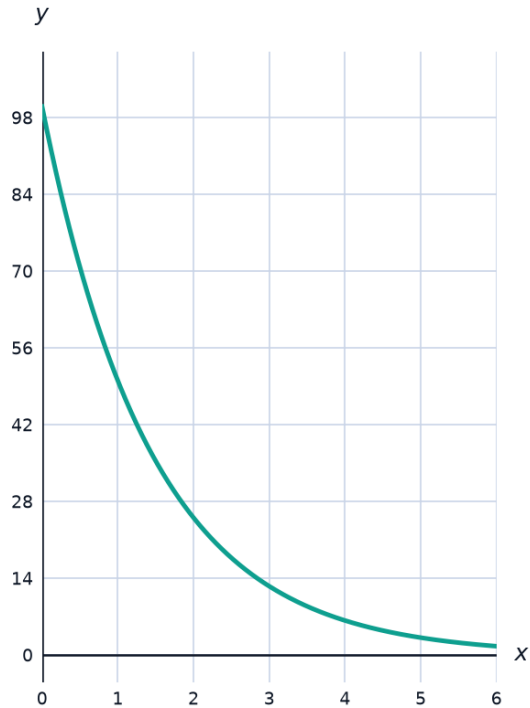
- 18 Find $g(9)$ for $g(x) = \sqrt{x} + 4$.

Exponential functions

- 19 If $f(x) = 2^x$, find $f(5)$ and $f(-2)$.

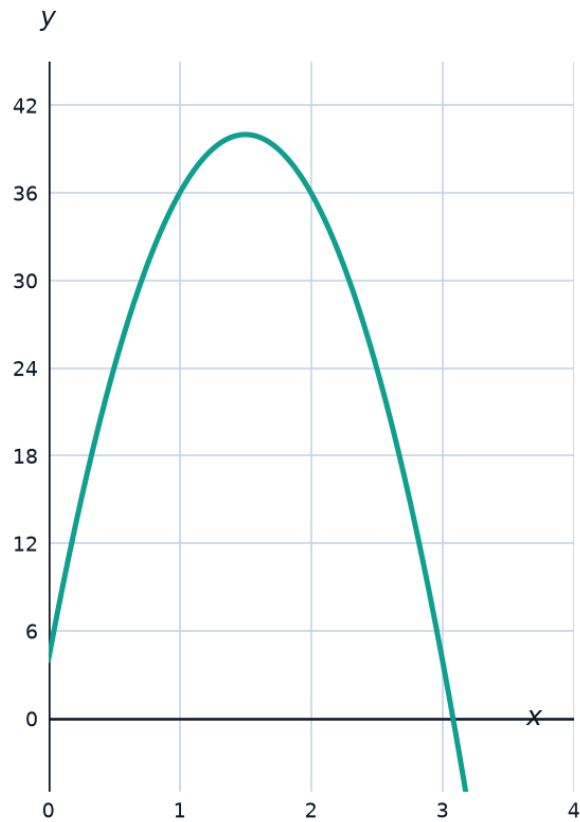


- 20 A function is given by $g(x) = 100(0.5)^x$. Find $g(3)$ and interpret its meaning if this models a decaying quantity over x time periods.



Multi-part analysis of a real-world function model

- 21** This question has two parts. A ball is thrown, and its height is modeled by $h(t) = -16t^2 + 48t + 4$, where t is time in seconds. (a) Find the height of the ball at $t = 0$ and interpret this value. (b) Find the time at which the ball reaches its maximum height, and find that maximum height.



Multiple choice — easy

- 22** If $f(x) = -2x^2 + 7x + 5$, find $f(-3)$.

- a. -34
- b. -36
- c. -32
- d. -44

23 If $f(x) = 4x^2 - 9x - 6$, find $f(1)$.

- a. -9
- b. -11
- c. -13
- d. 1

24 If $f(x) = -2x^2 + 4x + 9$, find $f(4)$.

- a. -5
- b. -9
- c. -25
- d. -7

25 If $f(x) = -3x^2 + 8x + 4$, find $f(6)$.

- a. -54
- b. -56
- c. -58
- d. -64

26 If $f(x) = 4x^2 + 9x - 4$, find $f(6)$.

- a. 202
- b. 194
- c. 196
- d. 192

27 If $f(x) = 2x^2 - 2x + 2$, find $f(-1)$.

a. 2

b. 6

c. 4

d. 8

28 If $f(x) = -3x^2 - 7x - 6$, find $f(1)$.

a. -4

b. -18

c. -14

d. -16

29 If $f(x) = 3x^2 + 3x - 1$, find $f(-1)$.

a. -1

b. 1

c. 4

d. -3

30 If $f(x) = 3x^2 + 5$, find $f(2)$.

a. 15

b. 7

c. 19

d. 17

31 If $f(x) = -x^2 - 9x + 4$, find $f(1)$.

- a. -4
- b. -14
- c. -6
- d. -8

32 If $f(x) = x^2 - 6x + 3$, find $f(2)$.

- a. -5
- b. -3
- c. -11
- d. -7

33 If $f(x) = -x^2 - 8x + 6$, find $f(-2)$.

- a. 20
- b. 18
- c. 6
- d. 16

34 If $f(x) = -x^2 + x + 8$, find $f(-4)$.

- a. -12
- b. -28
- c. -10
- d. -14

35 If $f(x) = 3x^2 - 9$, find $f(-3)$.

a. 36

b. 18

c. 20

d. 16

36 If $f(x) = x^2 - 5x - 6$, find $f(-4)$.

a. 42

b. 32

c. 30

d. 28

Multiple choice — medium

37 Find the vertex of $y = -3x^2 + 42x - 134$.

a. (7, 13)

b. (7, -13)

c. (-7, 13)

d. (8, 13)

38 Find the vertex of $y = 3x^2 + 36x + 107$.

a. (-5, -1)

b. (6, -1)

c. (-6, 1)

d. (-6, -1)

39 Find the vertex of $y = 3x^2 - 42x + 141$.

a. (7, 6)

b. (8, -6)

c. (7, -6)

d. (-7, -6)

40 Find the vertex of $y = -2x^2 - 4x + 4$.

a. (-1, -6)

b. (0, 6)

c. (-1, 6)

d. (1, 6)

41 Find the vertex of $y = -3x^2 + 12x - 20$.

a. (3, -8)

b. (2, -8)

c. (2, 8)

d. (-2, -8)

42 Find the vertex of $y = x^2 - 14x + 46$.

a. (7, 3)

b. (8, -3)

c. (-7, -3)

d. (7, -3)

43 Find the vertex of $y = -x^2 + 16x - 60$.

a. (9, 4)

b. (8, 4)

c. (-8, 4)

d. (8, -4)

44 Find the vertex of $y = -x^2 - 2x - 13$.

a. (0, -12)

b. (-1, 12)

c. (-1, -12)

d. (1, -12)

45 Find the vertex of $y = 3x^2 + 48x + 195$.

a. (8, 3)

b. (-8, -3)

c. (-8, 3)

d. (-7, 3)

46 Find the vertex of $y = -3x^2 - 42x - 161$.

a. (-7, -14)

b. (-7, 14)

c. (7, -14)

d. (-6, -14)

47 Find the vertex of $y = -3x^2 - 6x + 9$.

a. $(-1, -12)$

b. $(-1, 12)$

c. $(1, 12)$

d. $(0, 12)$

48 Find the vertex of $y = -x^2 - 8x - 3$.

a. $(4, 13)$

b. $(-3, 13)$

c. $(-4, -13)$

d. $(-4, 13)$

49 Find the vertex of $y = -3x^2 + 42x - 136$.

a. $(7, -11)$

b. $(7, 11)$

c. $(-7, 11)$

d. $(8, 11)$

50 Find the vertex of $y = 2x^2 + 16x + 45$.

a. $(-4, -13)$

b. $(-4, 13)$

c. $(-3, 13)$

d. $(4, 13)$

51 Find the vertex of $y = -x^2 - 2x - 1$.

- a. $(1, 0)$
- b. $(-3, 1)$
- c. $(0, 0)$
- d. $(-1, 0)$

Multiple choice — hard

52 If $f(x) = -x + 4$ and $g(x) = 3x + 9$, find $f(g(0))$.

- a. 9
- b. -7
- c. -5
- d. -3

53 If $f(x) = 4x + 5$ and $g(x) = 2x + 6$, find $f(g(4))$.

- a. 63
- b. 61
- c. 59
- d. 14

54 If $f(x) = -x - 8$ and $g(x) = -3x + 1$, find $f(g(2))$.

- a. -3
- b. -7
- c. -5
- d. -1

55 If $f(x) = 2x - 4$ and $g(x) = -3x - 5$, find $f(g(0))$.

a. -14

b. -5

c. -16

d. -12

56 If $f(x) = -4x + 8$ and $g(x) = -x + 1$, find $f(g(-1))$.

a. -2

b. -3

c. 0

d. 2

57 Find the inverse of $f(x) = 3x + 9$.

a. $\frac{x+9}{3}$

b. $\frac{x-7}{3}$

c. $\frac{x-9}{3}$

d. $\frac{x-11}{3}$

58 Find the inverse of $f(x) = -5x + 11$.

a. $\frac{x-9}{-5}$

b. $\frac{x-11}{-5}$

c. $\frac{x+11}{-5}$

d. $\frac{x-13}{-5}$

59 Find the inverse of $f(x) = 4x + 11$.

- a. $\frac{x-9}{4}$
- b. $\frac{x+11}{4}$
- c. $\frac{x-13}{4}$
- d. $\frac{x-11}{4}$

60 Find the inverse of $f(x) = 2x - 1$.

- a. $\frac{x+1}{2}$
- b. $\frac{x-1}{2}$
- c. $\frac{x+3}{2}$
- d. $\frac{x+1}{3}$

61 Find the inverse of $f(x) = 5x - 3$.

- a. $\frac{x+5}{5}$
- b. $\frac{x-3}{5}$
- c. $\frac{x+1}{5}$
- d. $\frac{x+3}{5}$

62 If $f(x) = 5 \cdot 3^x$, find $f(2)$.

- a. 30
- b. 48
- c. 45
- d. 42

63 If $f(x) = 2 \cdot 3^x$, find $f(3)$.

a. 18

b. 54

c. 57

d. 51

64 If $f(x) = 2 \cdot 4^x$, find $f(4)$.

a. 516

b. 508

c. 32

d. 512

65 If $f(x) = 4 \cdot 4^x$, find $f(-2)$.

a. $\frac{1}{8}$

b. 1

c. -32

d. $\frac{1}{4}$

66 If $f(x) = 4 \cdot 4^x$, find $f(3)$.

a. 252

b. 48

c. 256

d. 260